

# Task 1: Research

*COS 221 – Team 27*

## 1. Overview

Online travel has grown into one of the largest e-commerce sectors, with platforms like Expedia, Booking.com, and TripAdvisor handling flights, hotels, and experiences in a single website. These websites rely on having a wide selection of travel options, and making prices clear and easy to compare [1]. The core problem they have not fully solved is trust: users often cannot verify whether a package is good value or whether an agency is reliable. Tripistry targets this gap by letting travellers compare packages from multiple agencies and leave verified reviews, while agencies get a controlled space to publish and manage their offerings.

## 2. Travel Product Categories

A travel package typically bundles four types of content:

- **Flights:** Priced dynamically based on demand and lead time. Aggregators like Skyscanner pull data from multiple carriers [1].
- **Accommodation:** Hotels, guesthouses, and self-catering units. Star ratings and guest reviews are the main quality signals [2].
- **Tourist attractions:** Museums, guided tours, parks. OpenTripMap provides open point-of-interest data useful for seeding a database [3].
- **Restaurants:** Often curated by destination guides or crowd-sourced via TripAdvisor and Google Maps.
- **Travel Agencies:** Traditional and online agencies range from large players like Flight Centre to smaller operators.
- **Popular destinations:** High-demand locations such as Paris, Dubai, Bali, Cape Town, and New York are typically categorised by region, season, and travel style (luxury, adventure, cultural tourism) to help users compare packages efficiently

## 3. Rating and Categorisation

Most platforms use a 1 to 5 star rating system as well as written reviews. Hu et al. (2017) found that the number of reviews matters almost as much as the average score, a product with many average reviews tends to convince people better than one with a small amount of excellent ones [2]. To prevent fake reviews, websites use purchase records or flag statistical outliers. For Tripistry, ratings apply for both agencies and packages, they are the primary basis on which travellers compare options. Packages are also typically filtered by price range (economy, mid-range, luxury), destination, and duration.

## 4. Group Trips, Reviews, and Recommendations

Solo travellers or small parties who want a structured trip but lack companions are mostly forgotten by standard booking platforms. Operators like Intrepid Travel and G Adventures fill this need with open-group departures that anyone can join. Tripistry includes this feature through agency-managed group trips. On the recommendations side, large platforms use collaborative filtering to suggest packages based on past behaviour [4], an easier method involves boosting higher rated packages in a searched destination. User reviews remain the biggest factor in travel decisions and should be prominently displayed on both agency and package pages.

## References

- [1] Expedia Group Media Solutions, “State of Travel 2023,” Expedia Group, 2023. [Online]. Available: <https://advertising.expedia.com/state-of-travel-2023>. [Accessed: 1 May 2026].
- [2] Y. Hu, J. Pavlou, and J. Zhang, “Overcoming the J-shaped distribution of product reviews,” *Communications of the ACM*, vol. 60, no. 5, pp. 144–153, 2017.
- [3] OpenTripMap, “OpenTripMap API Documentation,” 2023. [Online]. Available: <https://opentripmap.io/docs>. [Accessed: 1 May 2026].
- [4] F. Ricci, L. Rokach, and B. Shapira, “Recommender Systems: Introduction and Challenges,” in *Recommender Systems Handbook*, 2nd ed. New York, NY: Springer, 2015, pp. 1–34.